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Keywords

- Login
- Register
- Logout
- Change Password
- Set Username / Change Username
- Upload Profile Picture
- Edit Basic Info
- Dashboard
- Attendance
- Task / Assignment
- Announcements
- Role System
- Smart Fees
- Search
- Notifications
- Leave Request
- Bookmark
- Dark Mode

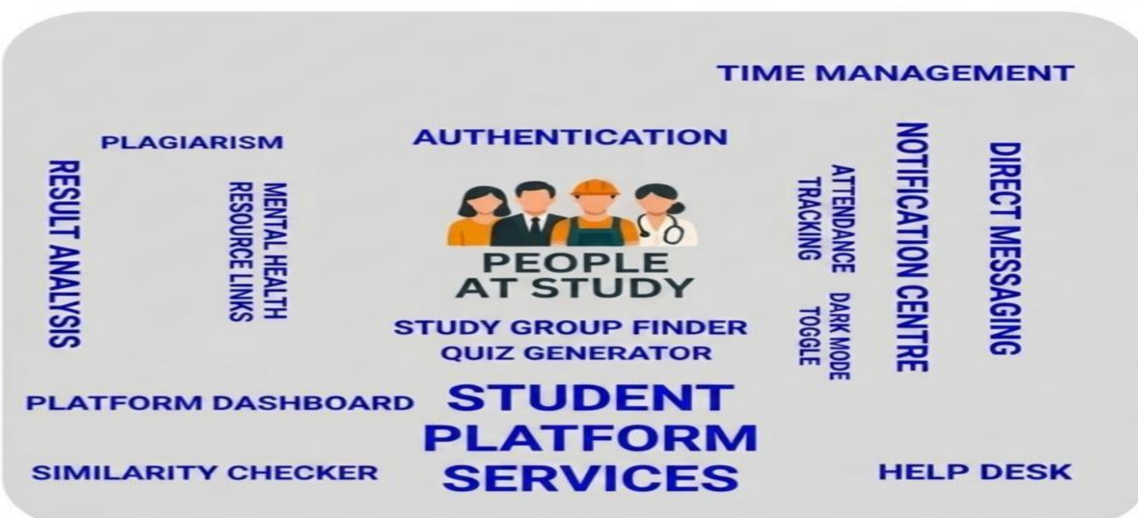

Figure 1-Keywords

Introduction

Students and teachers in schools and colleges in Nepal face a lot of problems when it comes to managing day-to-day activities. Things like tracking attendance, submitting assignments, checking fee status, and getting announcements are all done manually or on different platforms which makes it confusing and slow. There is no single place where a student can log in and see everything they need.

To solve this problem, we came up with Abhyas, which means "practice" or "study" in Nepali. Abhyas is a web-based education management system that is made for students and teachers. It brings together all the important features of a school or college management system into one easyto-use platform. From marking attendance to submitting assignments to checking if fees are paid, everything can be done from one place.

Abhyas is designed to be simple and straightforward. We did not want to make something overly complicated. We wanted to build something that a regular student or teacher can open and understand without needing any training. The system also has a role-based structure, meaning teachers and students see different things based on who they are.

Figure 2- Abhyas Logo

Aims

The main aim of Abhyas is to give students and teachers one platform where they can manage all education-related activities easily and without confusion. We want to remove the need for manual processes like paper attendance, physical notice boards, and separate fee tracking systems by putting everything in one digital system that is accessible from anywhere.

Objectives

- To develop a secure authentication system with login, registration, and password management features.
- To create profile management features for students and teachers.
- To design a centralized dashboard for tasks, attendance, and announcements.
- To implement an attendance management system for recording and viewing attendance.
- To develop an assignment submission system for teachers and students.
- To provide an announcement system for sharing important information.
- To implement role-based access control for different users.
- To develop a fee tracking system for monitoring payment status.
- To improve usability through search, notifications, leave requests, and bookmarks.
- To include additional features such as dark mode, chat, and basic AI tools if feasible.

Problem Statement

In most schools and colleges in Nepal, managing student and teacher activities is still done the old way. Attendance is taken on paper. Assignments are handed over physically or through random messaging apps like WhatsApp. Announcements are put on a notice board that not everyone sees. Fee records are kept in registers that can get lost or damaged. There is no single system that connects all of these together in one place.

This causes a lot of problems for both students and teachers. Students sometimes miss assignments because they did not see them in time. Teachers have to manually track who is present and who is not. Admins have to check paper records to know who has paid fees and who has not. Everything takes more time and effort than it needs to.

Abhyas is built to fix these problems. Instead of having everything scattered across different places, Abhyas puts everything together in one web-based system that is easy to use and accessible from any device. Students can log in, check their tasks, mark attendance, see announcements, and check their fee status all in one place. Teachers can manage tasks, post announcements, and approve leave requests. Admins can manage fees and users. This makes the whole process faster, cleaner, and easier to manage.

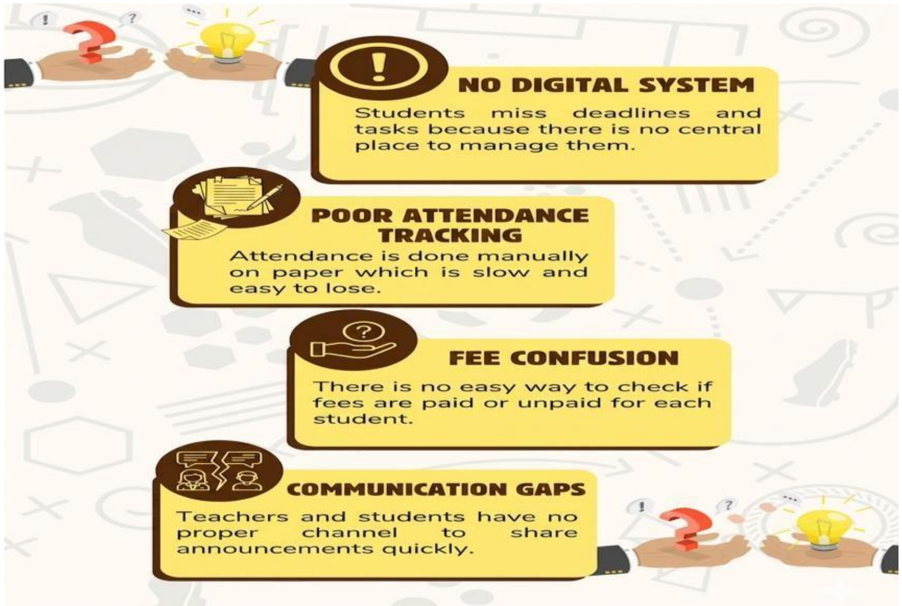

Figure 3-Problem Statement

Literature Review

The need for digital education management tools has been growing, especially after the COVID19 period when many schools and colleges had to go online for a short time. This showed that most institutions in Nepal were not ready for digital management. While some global platforms like Google Classroom and Microsoft Teams exist, they are not fully suited to the specific needs of Nepali schools and colleges. They are either too complex or lack features that are needed locally, like fee tracking or Nepali-context attendance systems.

Research has shown that simple and well-designed platforms are more likely to be used regularly by students and teachers compared to complicated systems with too many options. A system that is focused on what users actually need every day is more effective than one that tries to do everything at once (AltexSoft Editorial Team, 2023).

Mobile-friendly and web-based systems are also becoming more important as more students use phones and laptops to access the internet. A system that works on any browser without needing to install anything is more practical for most students in Nepal (Joshi, 2024).

Abhyas is built keeping all of this in mind. It focuses on the daily needs of students and teachers, keeps the interface simple, and works on any browser. It is not trying to compete with global giants like Google Classroom but instead targets the very specific and practical problems that students and teachers in Nepal face every day.

Features

- Authentication - Register, Login, Logout, Change Password
- Profile - Set Username, Change Username, Upload Profile Picture, Edit Basic Info
- Dashboard - View Today's Tasks, Attendance Status, Announcements
- Attendance - Mark Present, Auto Absent, View History
- Task / Assignment - Add Task, Submit, Re-upload, View Status
- Announcements - Post and View Announcements
- Role System - Teacher Role, Student Role
- Smart Fees - Mark Paid, Mark Unpaid, View Fee Status
- Search - Search Tasks, Search Announcements
- Notifications - New Task Alert, Deadline Reminder
- Leave Request - Student Applies, Teacher Approves or Rejects
- Bookmark - Save Task as Bookmark
- Dark Mode - Toggle Light and Dark Theme
- Simple Chat - Student Messages Teacher, Teacher Replies
- Basic AI - Generate Quiz from Text, Explain a Topic

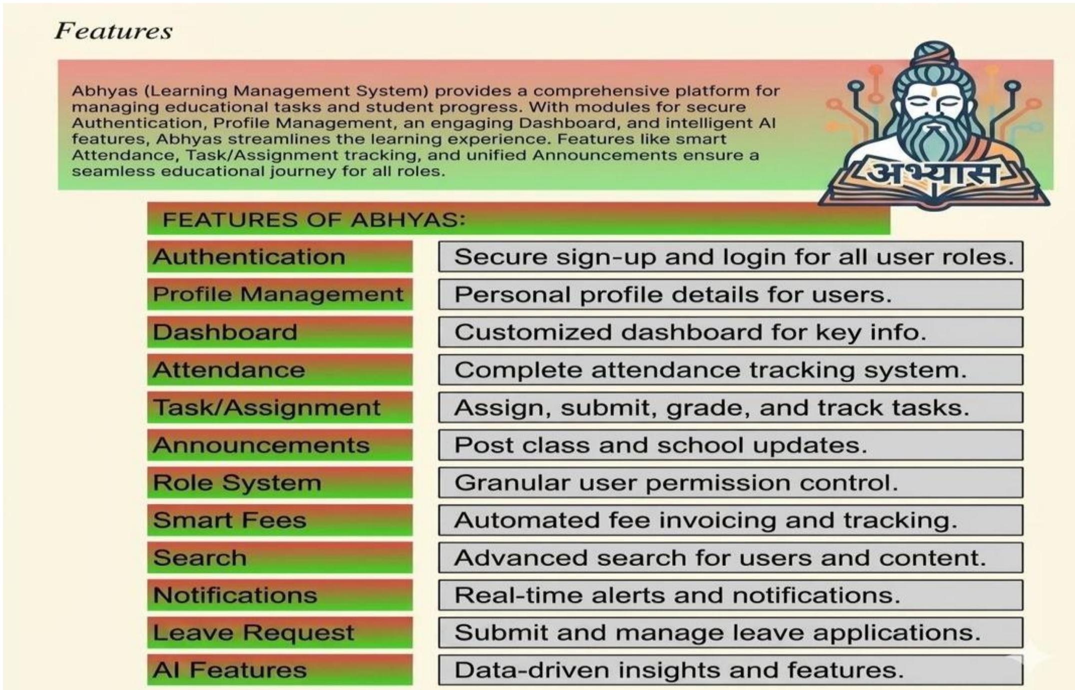

Figure 4- Features of Abhyas

Functionalities

Functional Requirements

Functional requirements describe what the system must do. These are the actual actions and features that the system needs to carry out so that users can do their work properly (AltexSoft Editorial Team, 2023).

Table 1 - Functional Requirement of Abhyas

Requirement ID	Functional Requirement	Description
FR-001	User Authentication	Lets users register an account, log in securely, log out, and change their password.
FR-002	Profile Management	Allows users to set or change username, upload a profile picture, and edit basic info.
FR-003	Dashboard	Shows today's tasks, attendance status, and latest announcements in one place.
FR-004	Attendance System	Users can mark present once per day. Auto absent runs at end of day. History is viewable.
FR-005	Task / Assignment System	Teachers post tasks, students submit and reupload before deadline. Status is tracked.
FR-006	Announcements	Teachers and admins post announcements. All students can view them.
FR-007	Role System	Teachers can manage tasks and attendance. Students can only view and submit.
FR-008	Smart Fees	Admin marks fees as paid or unpaid. Students can see their fee status.
FR-009	Search	Users can search tasks or announcements by title or keyword.
FR-010	Notifications	Students get alerts for new tasks and deadline reminders.
FR-011	Leave Request	Students apply for leave. Teachers approve or reject the request.
FR-012	Bookmark	Students can save important tasks to their bookmarks list.

Non Functional Requirements

Non-functional requirements describe how the system should behave. They are about performance, security, and reliability rather than specific features.

Table 2- ***Non Functional Requirements of Abhyas***

Requirement ID	Non Functional Requirement	Description
NFR-001	Performance	Pages should load within 2 to 3 seconds even with multiple users active at the same time.
NFR-002	Usability	The interface should be simple enough for any student or teacher to use without training.
NFR-003	Security	All user data is protected. Passwords are stored in encrypted form. Sessions are secure.
NFR-004	Scalability	The system should handle more users and data as the number of students grows over time.
NFR-005	Availability	The system should be available at all times and work on any browser or device.

Development Methodology

We have chosen the Agile model for developing Abhyas. Agile is a way of building software step by step instead of trying to build everything at once. It lets us get feedback early, fix problems quickly, and adjust the plan if something changes. This is a good fit for our project because we are building it as part of a semester, which means we need to manage our time carefully and keep making progress in small stages (Gurnov, 2025).

Framework

We will use the Scrum framework as our specific approach within Agile. Scrum works by dividing the work into short periods called Sprints, usually one to two weeks long. At the start of each sprint, we plan what we are going to build. Then we work on it and check the results at the end. This helps us stay on track and know exactly where we are at all times (Malsam, 2024).

We will hold short daily check-ins to talk about what we did, what we plan to do next, and if anything is blocking us. At the end of each sprint, we review the work and discuss what went well and what can be improved. This cycle keeps the development process moving forward in a healthy way.

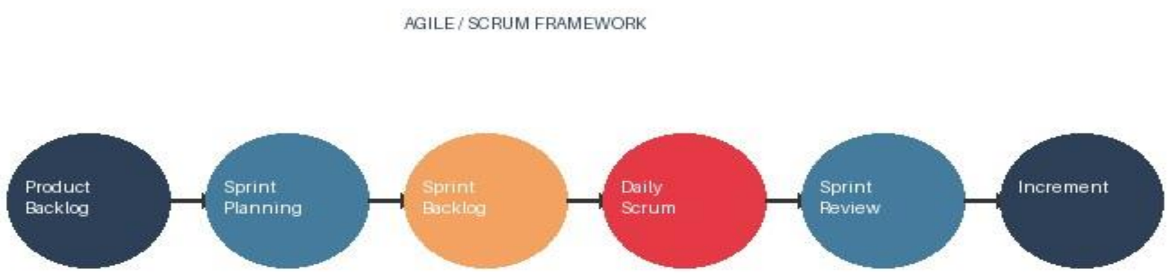

Figure 5- Agile Scrum Framework

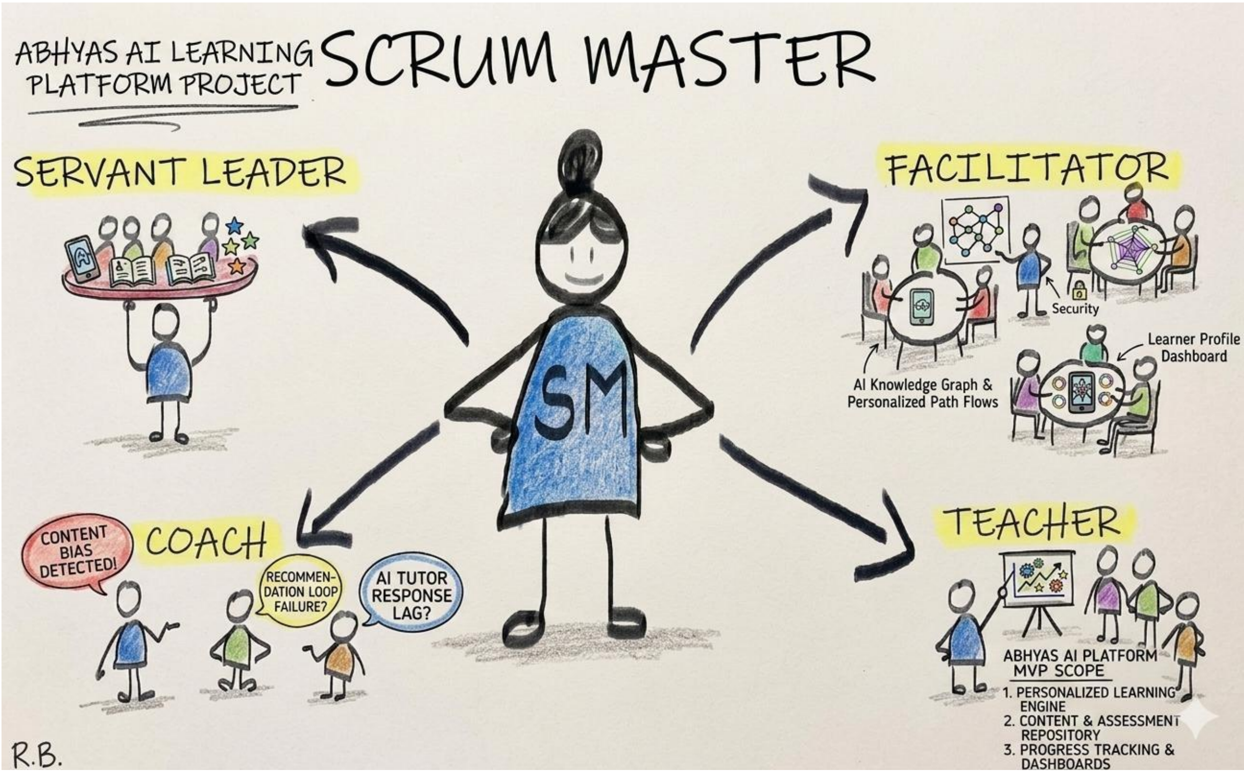

Figure 6- Scrum Master

The Scrum Master in our team is responsible for making sure the team follows the Scrum process. They help remove any blockers that slow the team down and make sure everyone is communicating well.

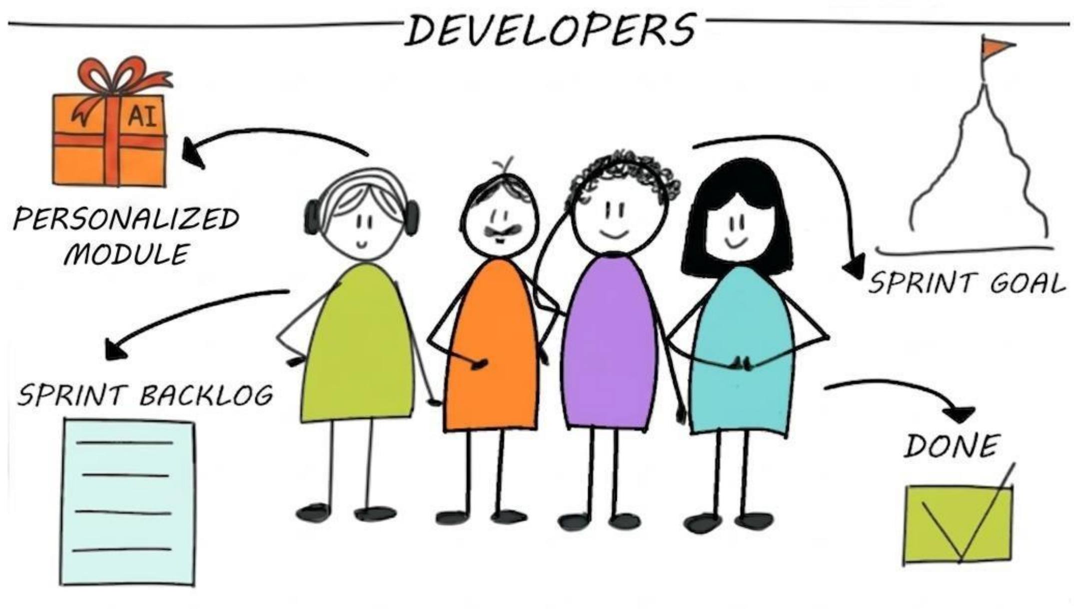

Figure 7- Developers

The developers in our team are responsible for actually building the features. Each sprint, developers pick tasks from the sprint backlog and work on completing them. They collaborate and help each other when something is difficult.

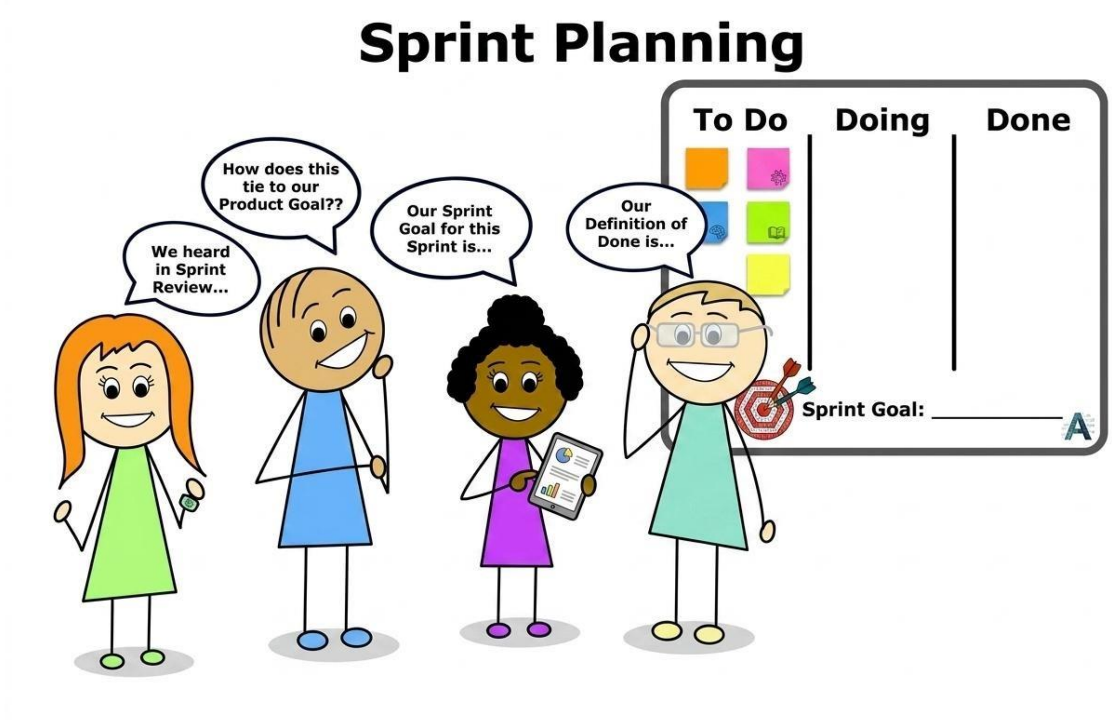

Figure 8- Sprint Plaining

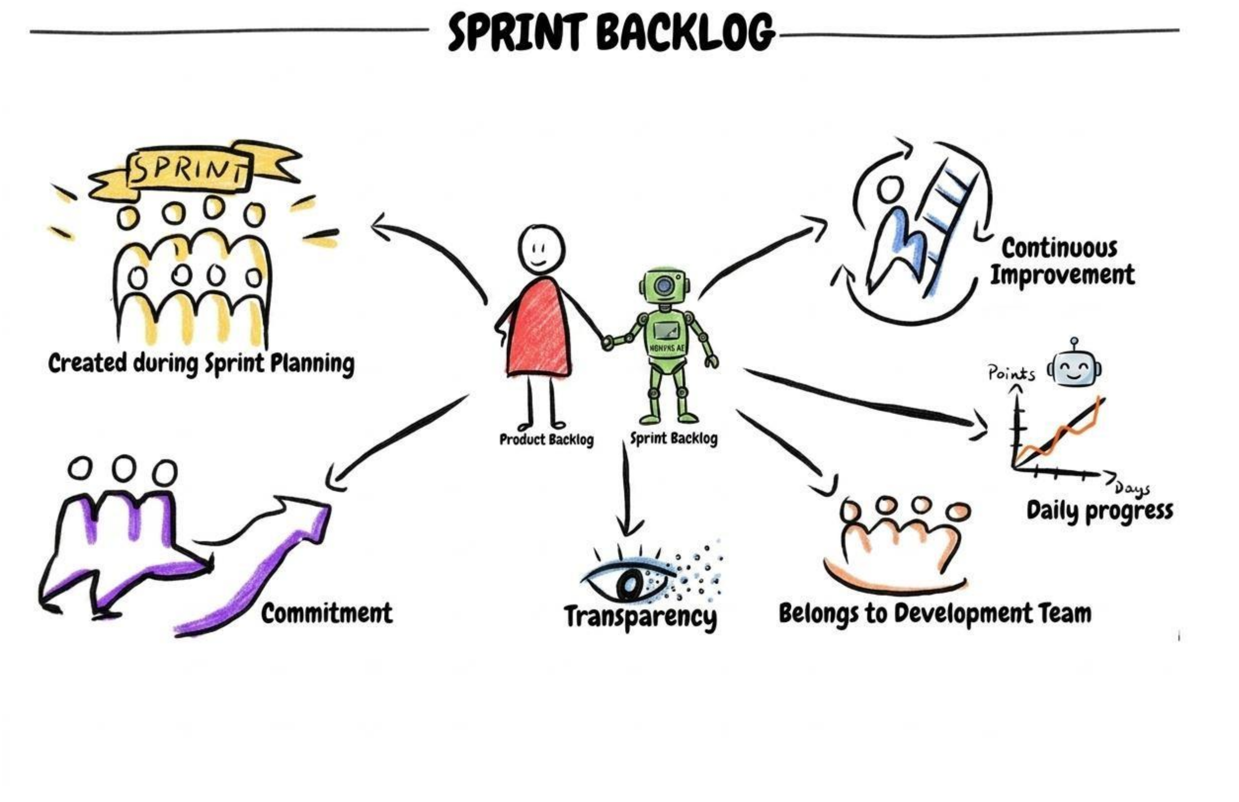

Figure 9- Sprint Backlog

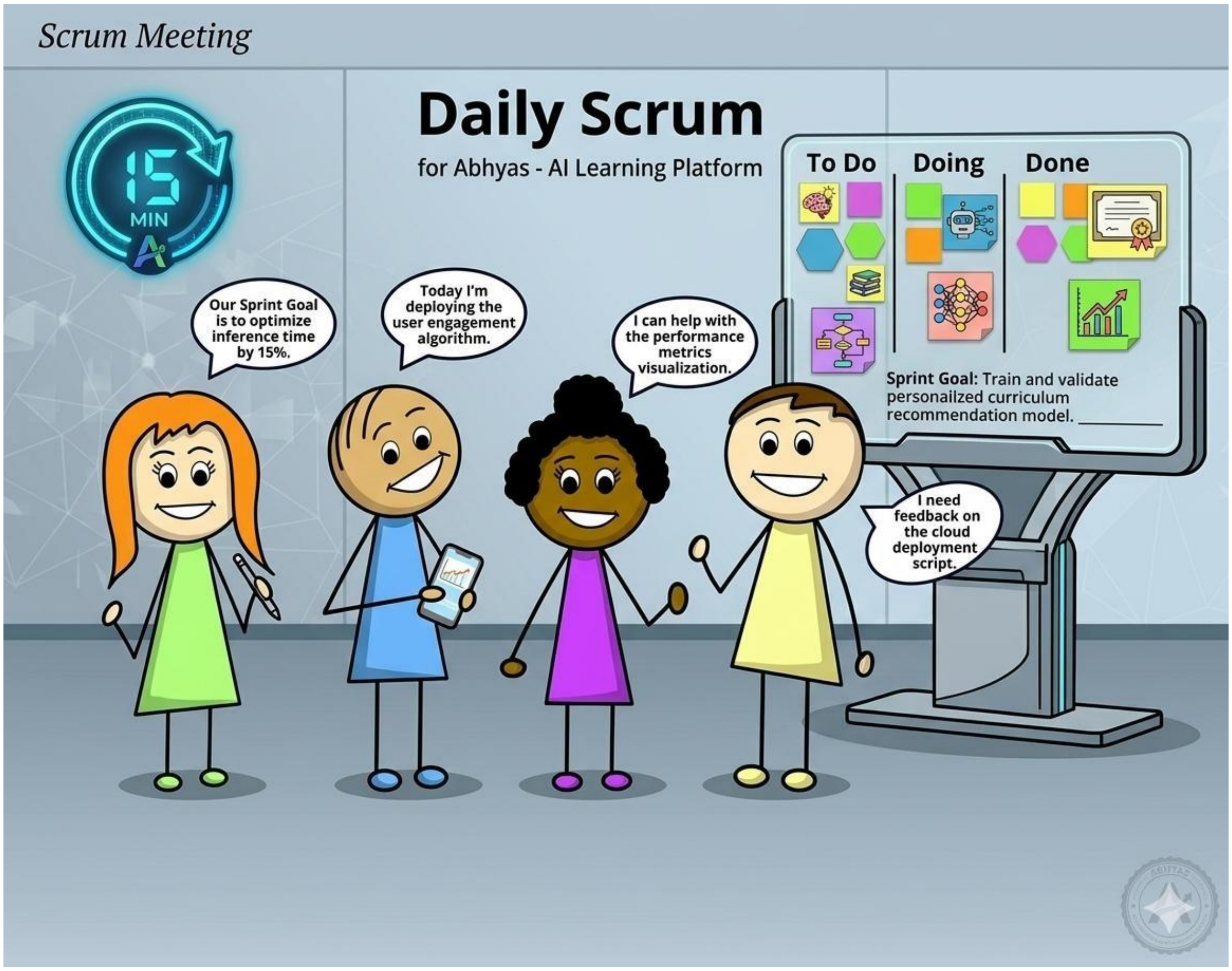

Figure 10- Scrum Meeting

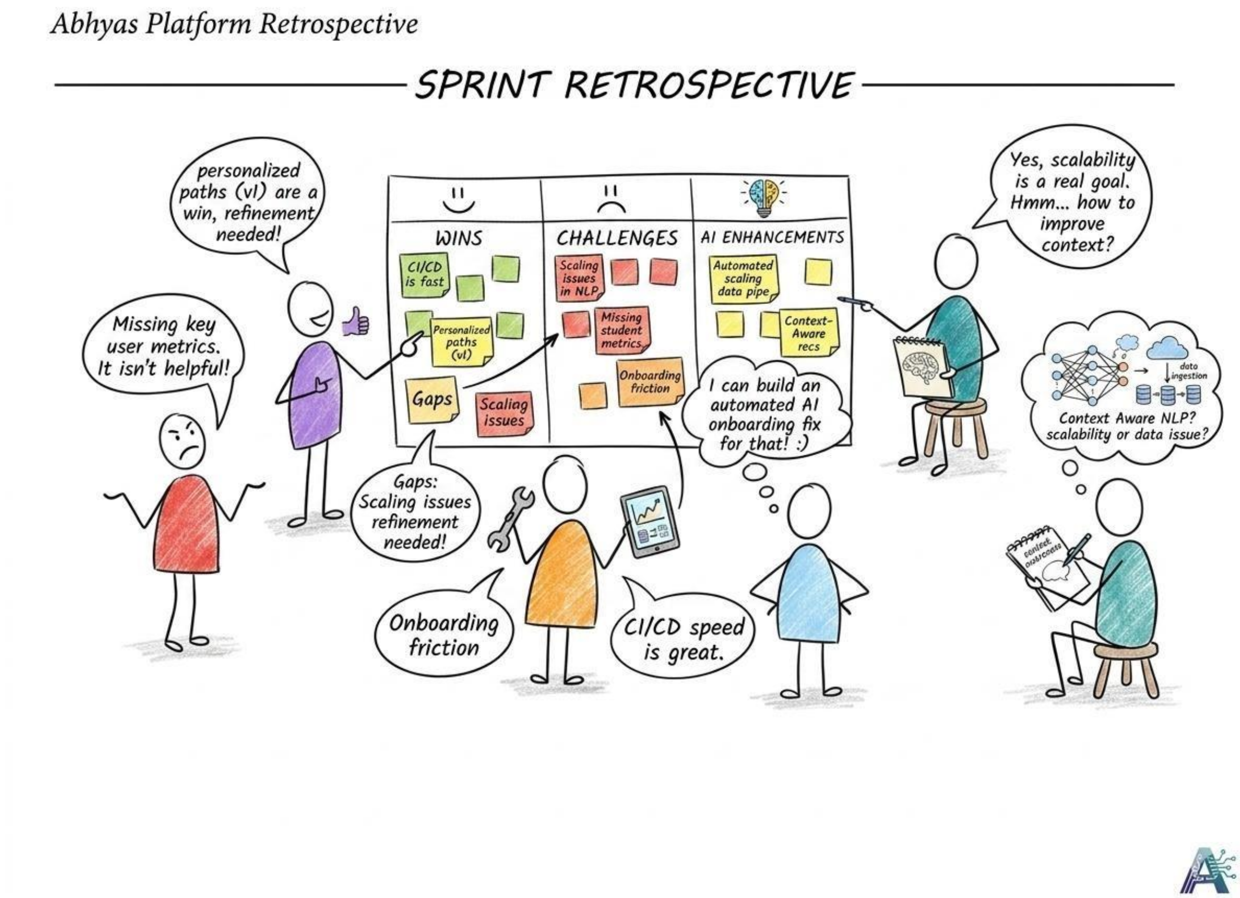

Figure 11- Scrum Retrospective

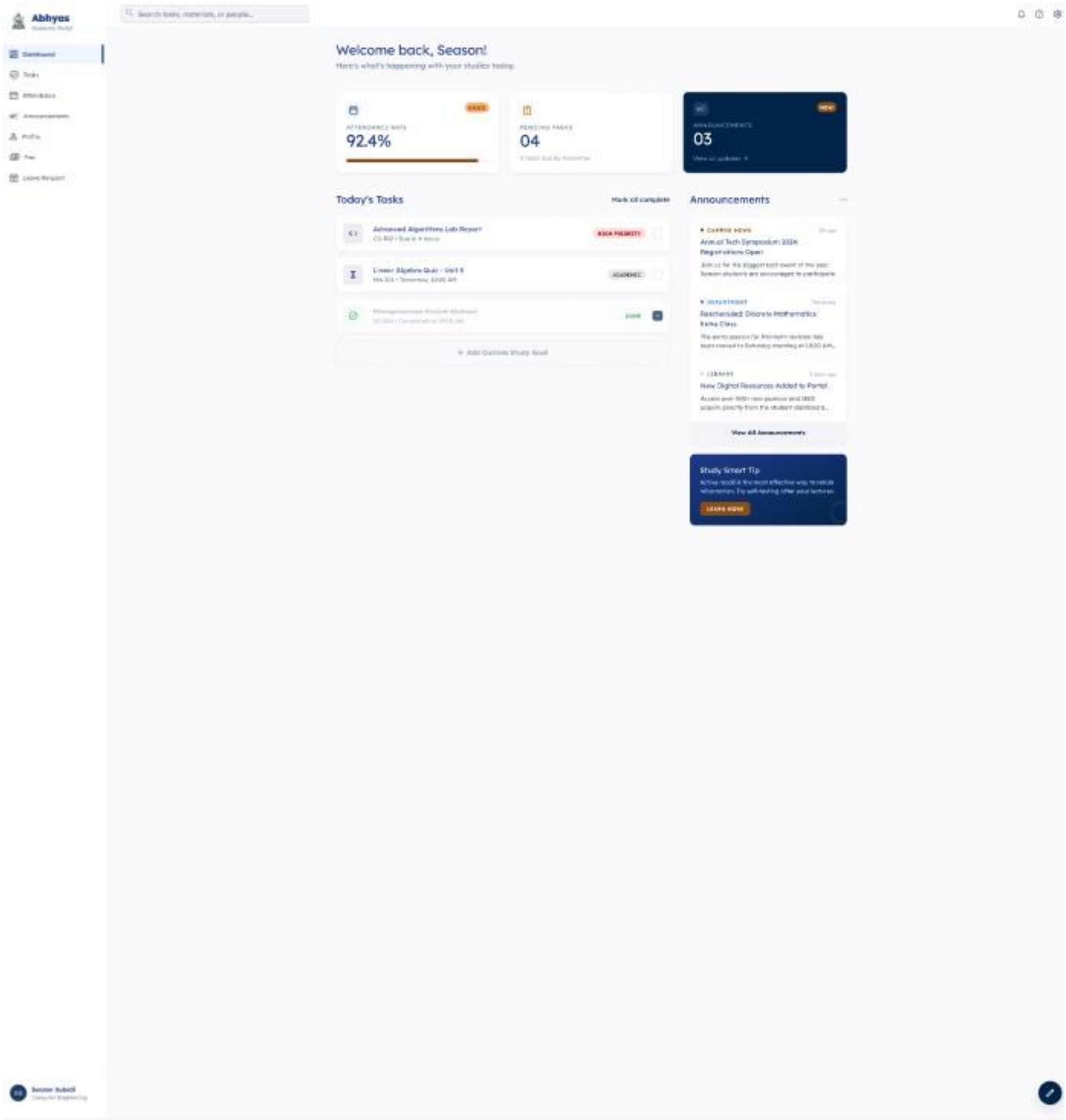

Figure 12-Sample Theme

Tools and Technologies

Here is a list of tools and technologies we will be using to build and manage the Abhyas project:

- Visual Studio Code as our main code editor
- Trello for managing tasks and keeping track of sprint progress
- Google Meet and WhatsApp for team communication and meetings
- Figma for designing the UI prototype before building it
- Plain HTML/CSS/JS for the front end
- Flask for the back end
- MySQL for the database
- Git and GitHub for version control and collaboration
- Draw.io for drawing system diagrams
- MS Word, MS Excel, and Google Docs for writing documentation and sprint records

Figure 13- Tool and Technique

Conceptual Diagram

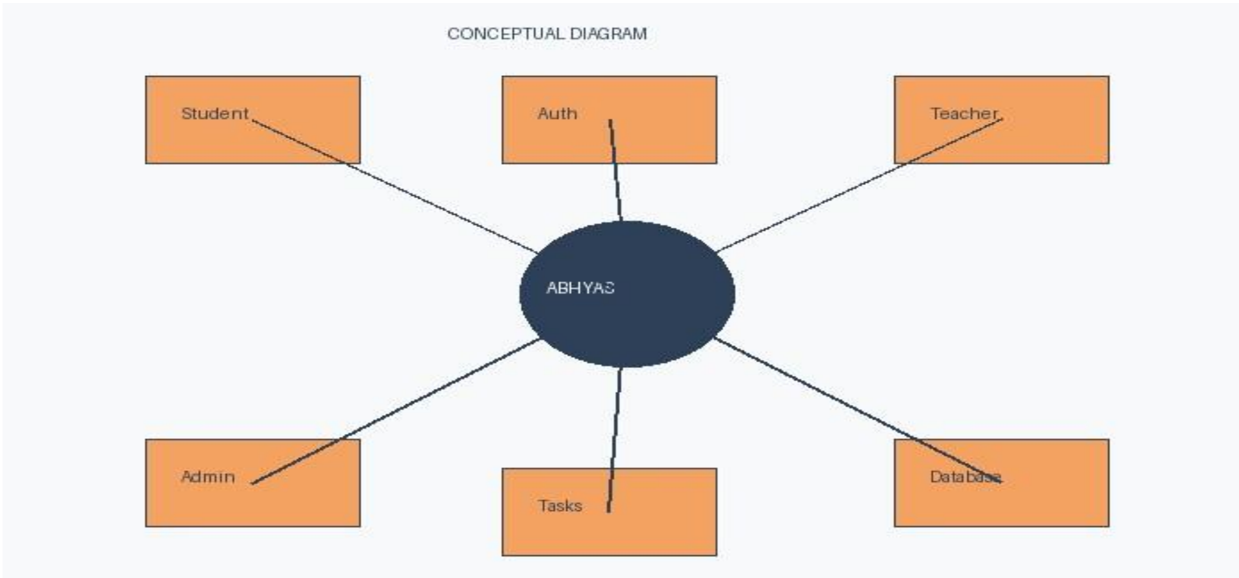

Figure 14- Conceptual Diagram of Abhyas

Scope

- Student and teacher account management
- Daily attendance tracking and history
- Assignment posting, submission, and deadline tracking
- Announcement board for teachers and admins
- Fee payment status tracking for students
- Role-based access for students, teachers, and admins
- Leave request management
- Search, notifications, and bookmark features
- Optional features: dark mode, simple chat, basic AI tools

SWOT Analysis

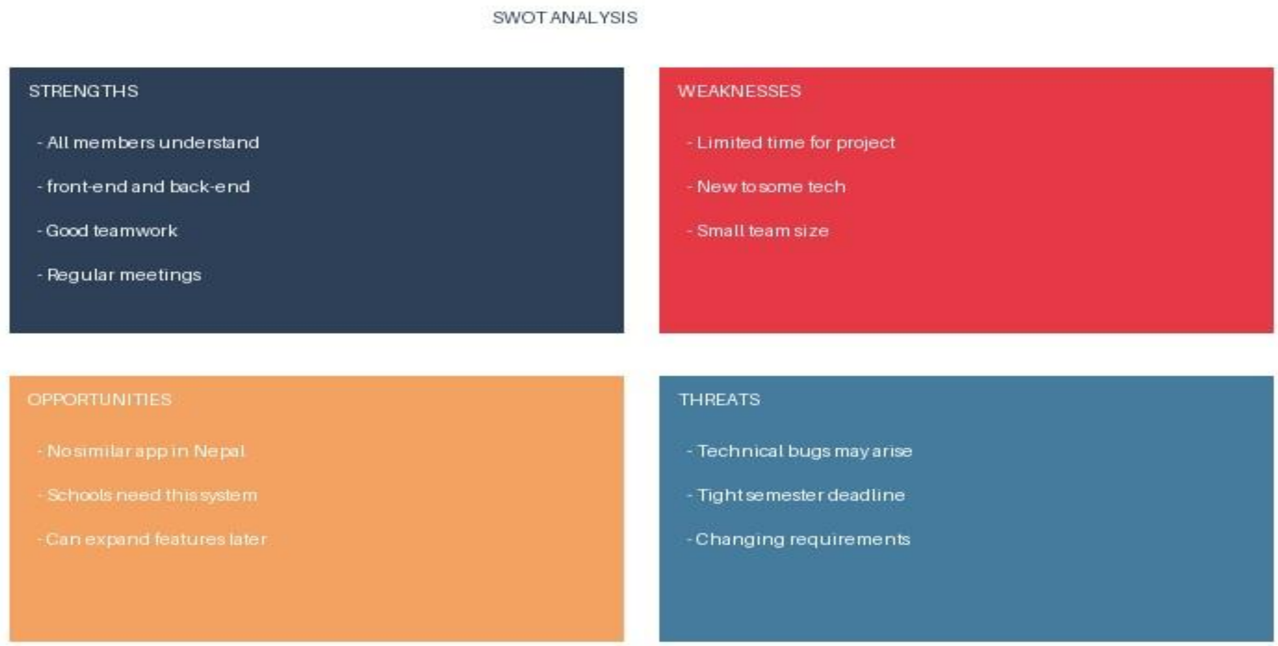

Figure 15- SWOT Analysis of Abhyas

Conclusion

Abhyas is a practical and straightforward education management system that is built to solve real problems faced by students and teachers in Nepal. By putting attendance, assignments, announcements, fees, and other tools into one web-based platform, Abhyas removes the confusion and inefficiency that comes from managing all of these things separately.

This proposal has described the goals, features, methodology, and scope of the project. The system is designed to be simple to use so that any student or teacher can get started without needing technical knowledge. The Agile and Scrum approach will help us build the system step by step while making sure we stay on track and deliver something that actually works.

Overall, Abhyas has the potential to genuinely improve the daily experience of students and teachers by giving them one reliable platform they can depend on. We are confident that with good teamwork and proper planning, we will be able to deliver a working system that meets the needs of our users.

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Appendix

Figma Design → https://www.figma.com/design/UVJv1w0Ci5IJaQa4wPSBBj/abhyas?node-id=60-6&t=GYQfBUGMOyOLdmjt-1

This includes Low fidelity and High Fidelity design